Note : Do all types of Problems taught in the classroom from different units.

OS queston Bank UNIt 1

Unit1

1. **Define Operating System with diagram.**
2. **State the major components of a computer system and their roles in system organization.**
3. **Identify the different types of system calls and provide examples for each type.**
4. **Distinguish between system calls and system programs.**
5. **Sketch and explain about the storage structure in detail.**
6. **Illustrate the different types of operating systems with examples.**
7. State the purpose of a bootstrap program
8. **Identify the main components of computer system organization.**
9. Discuss the services provided by an operating system and their significance to users and applications.
10. Analyse the characteristics of various types of operating systems, including real-time, batch, and time-sharing systems

4-Mark Questions

1. Explain the system view of an operating system. *(L2)*
2. List the different storage structures used in operating systems. *(L2)*
3. Describe a monolithic operating system structure. *(L2)*
4. What is a layered operating system structure? *(L2)*
5. Explain the role of command interpreters in an operating system. *(L2)*

7-Mark Questions

1. Compare and contrast the user view and system view of an operating system. *(L3)*
2. Discuss how computer system organization affects the performance of an operating system. *(L3)*
3. Analyze the function of a bootstrap program in the system startup process. *(L3)*
4. Examine the advantages and disadvantages of a microkernel structure. *(L3)*
5. Describe the services provided by an operating system and their significance to users and applications. *(L3)*

12-Mark Questions

1. Discuss the concept of operating systems and their role in managing computer resources. *(L4)*
2. Evaluate the different types of storage structures used in operating systems and their advantages. *(L4)*
3. Analyze the characteristics of various types of operating systems, including real-time, batch, and time-sharing systems. *(L4)*

4. Examine the layered operating system structure and its impact on system design and implementation. *(L4)*
5. Evaluate the differences between command interpreters and graphical user interfaces in operating systems. *(L4)*

Unit 2

2-Mark Questions

1. Define a process.
2. What is the role of a Process Control Block (PCB)?
3. Identify the different states of a process.
4. What does context switching mean?
5. List the types of threads.

4-Mark Questions

1. Explain the concept of a process in memory.
2. Describe the process state transitions.
3. Discuss the operation on processes.
4. What are the benefits of using threads in programming?
5. Define the CPU–I/O burst cycle.

7-Mark Questions

1. Compare and contrast pre-emptive and non-preemptive scheduling.
2. Analyze the different states of a thread and their significance.
3. Examine the objectives of process scheduling.
4. Describe the types of CPU schedulers.
5. Discuss the criteria used for scheduling decisions.

12-Mark Questions

1. Discuss the overall concept of processes, including their lifecycle and management in an operating system. *(L4)*
2. Evaluate the advantages and disadvantages of using threads versus processes. *(L4)*
3. Analyze the scheduling algorithms, focusing on their principles and applications in process management. *(L4)*
4. Compare the First-Come, First-Served (FCFS) and Shortest Remaining Time First (SRTF) scheduling algorithms, highlighting their performance characteristics. *(L4)*

5. Examine the priority scheduling algorithm, including its advantages, disadvantages, and how it handles priority inversion. *(L4)*

Unit 3

2-Mark Questions

1. Define the critical section problem.
2. What is a race condition?
3. Describe Peterson's solution.
4. What is a semaphore?
5. Name the classic problems in interprocess communication (IPC).

4-Mark Questions

1. Explain the importance of synchronization in operating systems.
2. Describe how semaphores can be used to solve the critical section problem.
3. What are the characteristics of the producer-consumer problem?
4. Explain the reader-writer problem and its challenges.
5. What is the dining philosophers problem and its significance in synchronization?

7-Mark Questions

1. Compare and contrast deadlock prevention and deadlock avoidance.
2. Discuss the methods for handling deadlocks.
3. Analyze the conditions necessary for a deadlock to occur.
4. Explain the Banker's algorithm and its application in deadlock avoidance.
5. Describe a scenario where a race condition might occur and how it can be mitigated.

12-Mark Questions

1. Discuss the critical section problem in detail, including potential solutions and their effectiveness.
2. Evaluate the producer-consumer problem, explaining various synchronization techniques used to resolve it.
3. Analyze the dining philosophers problem, outlining its implications in concurrent programming and potential solutions.
4. Explain deadlock characterization and the implications for operating system design and management.
5. Discuss deadlock detection and recovery strategies, highlighting their advantages and disadvantages.

These questions cover a range of c

Unit4

2-Mark Questions (L1)

1. Define memory management in operating systems.
2. What is the difference between logical and physical addresses?
3. Explain the concept of swapping in memory management.
4. Define contiguous memory allocation.
5. What is internal fragmentation?

4-Mark Questions (L2)

1. Describe the mapping of logical addresses to physical addresses.
2. Explain the concepts of fixed partitioning and variable partitioning in memory allocation.
3. Discuss external fragmentation and its impact on memory allocation.
4. What is paging, and how does it help in memory management?
5. Explain the role of the Translation Lookaside Buffer (TLB) in paging.

7-Mark Questions (L3)

1. Analyze the advantages and disadvantages of contiguous memory allocation compared to paging.
2. Discuss the paging model for logical and physical memory with an example.
3. Describe the hardware requirements for implementing paging in an operating system.
4. Compare and contrast internal and external fragmentation, including their causes and solutions.
5. Explain how the TLB works and its significance in speeding up memory access.

12-Mark Questions (L4)

1. Discuss the basic concepts of memory management, including various allocation strategies and their implications on system performance. Include examples to illustrate your points. *(L4)*
2. Evaluate the paging technique in memory management, detailing its model, hardware requirements, advantages, and disadvantages. Discuss scenarios where paging is particularly beneficial. *(L4)*
3. Analyze how fragmentation (both internal and external) affects memory allocation efficiency, and discuss strategies to minimize fragmentation in a memory management system. *(L4)*

Unit no 5

2-Mark Questions (L1)

1. Define virtual memory and its purpose in an operating system.
2. What is demand paging?
3. Explain the concept of a page replacement algorithm.
4. Describe the structure of a disk.
5. What does FCFS stand for in disk scheduling?

4-Mark Questions (L2)

1. Discuss the advantages of using virtual memory.
2. Explain how demand paging improves memory utilization.
3. Describe the FIFO page replacement algorithm and its basic principle.
4. What are the key differences between LRU and Optimal page replacement algorithms?
5. Explain the SCAN disk scheduling algorithm.

7-Mark Questions (L3)

1. Analyze the working of the LRU page replacement algorithm and its implementation challenges.
2. Discuss the impact of page faults on system performance and how they can be minimized.
3. Compare and contrast different page replacement algorithms (FIFO, LRU, Optimal) in terms of efficiency and overhead.
4. Explain the C-SCAN disk scheduling algorithm and how it differs from the SCAN algorithm.
5. Evaluate the advantages and disadvantages of using virtual memory in modern operating systems.

12-Mark Questions (L4)

1. Discuss the complete mechanism of virtual memory management, including the concepts of paging, page tables, and page replacement algorithms. Provide a detailed example to illustrate your explanation. *(L4)*
2. Analyze various disk scheduling algorithms (FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK), detailing their advantages, disadvantages, and suitability for different types of workloads. Include examples to support your analysis. *(L4)*
3. Evaluate the impact of disk management on overall system performance, considering factors such as access time, throughput, and disk utilization. Discuss how different scheduling algorithms affect these factors. *(L4)*
4. Explain how virtual memory can lead to issues such as thrashing and discuss strategies to mitigate this problem. *(L4)*
5. Discuss the interaction between virtual memory management and disk scheduling in terms of overall system efficiency and performance. Provide examples of how these systems work together in real-world applications. *(L4)*

CPU SCHEDULing Problems

1. For the following set of processes, evaluate the effectiveness of FCFS, SJF, and Round Robin (quantum = 3) algorithms in terms of average waiting time and turnaround time.

- Process P1: Arrival = 0, Burst = 10
- Process P2: Arrival = 1, Burst = 5
- Process P3: Arrival = 2, Burst = 8
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2. Given the processes below, calculate the waiting and turnaround times using Priority Scheduling.

- Process A: Arrival = 0, Burst = 4, Priority = 1
- Process B: Arrival = 1, Burst = 3, Priority = 3
- Process C: Arrival = 2, Burst = 5, Priority = 2

Calculate the **Completion Time (CT)**, **Turnaround Time (TAT)**, and **Waiting Time (WT)** for each process using the **Shortest Job First (SJF)** non-preemptive scheduling algorithm with the following data:

Process	Arrival Time (AT)	Burst Time (BT)
A	0	7
B	1	4
C	2	1
D	3	5
E	4	2

Use the same formulas as before:

- **Turnaround Time (TAT)** = CT−AT\text{CT} - \text{AT}CT−AT
- **Waiting Time (WT)** = TAT−BT\text{TAT} - \text{BT}TAT−BT

Question 4: Priority Scheduling - Preemptive

Calculate the **Completion Time (CT)**, **Turnaround Time (TAT)**, and **Waiting Time (WT)** for each process using the **Priority Scheduling** algorithm with **preemption** based on the priority values (lower values indicate higher priority).

Process	Arrival Time (AT)	Burst Time (BT)	Priority
P1	0	4	2
P2	1	6	1
P3	2	2	4
P4	3	5	3

Disk Scheduling

Problem: Calculate the **total head movement** using the **First-Come, First-Served (FCFS)** algorithm for the following disk requests. The initial position of the disk head is at **100**.

Request Sequence

98, 183, 37, 122, 14, 124, 65, 67

Question 2: SSTF Disk Scheduling

Problem: Using the **Shortest Seek Time First (SSTF)** algorithm, calculate the total head movement for the following disk requests. The initial position of the disk head is at **50**.

Request Sequence

82, 170, 43, 140, 24, 16, 190

Question Set 3: Using the **SCAN** (Elevator) scheduling algorithm, calculate the **total head movement** for the following request sequence. Assume the disk head starts at **50** and moves toward **200**. The disk has cylinders numbered from **0** to **199**.

Request Sequence

82, 170, 43, 140, 24, 16, 190

Page replacement algorithm

Using the **First-In-First-Out (FIFO)** page replacement algorithm, calculate the **number of page faults** for the following reference string with **3 frames** available.

Reference String: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2

Question 2: Optimal Page Replacement

Calculate the **number of page faults** using the **Optimal Page Replacement** algorithm with **4 frames** for the following reference string:

Reference String: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

Question 3: LRU Page Replacement

Using the **Least Recently Used (LRU)** page replacement algorithm, find the **number of page faults** for the following reference string with **3 frames**.

Reference String: 3, 2, 1, 0, 3, 2, 4, 3, 2, 1, 0, 4

Question Bank 5: FIFO Page Replacement with More Frames

Using **FIFO Page Replacement**, determine the **number of page faults** when there are **4 frames** available for the following reference string:

Reference String: 2, 3, 4, 2, 1, 5, 2, 4, 5, 3, 2, 1, 4.